Climate Risk Scenario Analysis

Transition Risk, Physical Risk, and TCFD-Aligned Reporting

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Abstract

An applied treatment of climate risk scenario analysis for financial portfolios, covering both transition risk and physical risk under the NGFS scenario framework. Climate change poses a dual threat to financial stability: transition risk from the repricing of carbon-intensive assets during the shift to a low-carbon economy, and physical risk from the direct economic damages of rising temperatures, extreme weather events, and chronic environmental degradation. This document develops both risk channels from first principles. We formalize transition risk through carbon cost propagation and stranded asset valuation; physical risk through the Nordhaus damage function and extreme value theory for acute climate events. These are combined into a unified Climate Value-at-Risk metric spanning multiple NGFS pathways. A Sobol sensitivity analysis identifies the dominant risk drivers, and TCFD-aligned reporting metrics — Weighted Average Carbon Intensity, financed emissions, and implied temperature rise — are derived for portfolio-level disclosure. We assume familiarity with basic portfolio theory, Value-at-Risk, and the extreme value theory foundations from Project 05.

Contents

1 Why Climate Risk Matters for Finance **3**
1.1 The Regulatory Shift: From Voluntary to Mandatory . . . 3
1.2 Double Materiality . . . 3
1.3 The Scale of Exposure . . . 4

2 The NGFS Scenario Framework **4**
2.1 The Six Pathways . . . 4
2.2 Shadow Carbon Price Trajectories . . . 5
2.3 Short-Term Scenarios (May 2025) . . . 5

3 Transition Risk Modeling **5**
3.1 Carbon Cost Propagation . . . 5
3.2 Sector Emission Intensities . . . 6
3.3 Equity Repricing Under Carbon Costs . . . 6
3.4 Stranded Assets . . . 6

4 Physical Risk Modeling **7**
4.1 The Nordhaus Damage Function . . . 7
4.2 Acute Physical Risk: Extreme Weather Events . . . 8
4.3 Chronic Physical Risk . . . 9
4.4 Sector Exposure to Physical Risk . . . 9

5 Climate Value-at-Risk **10**
5.1 Definition . . . 10
5.2 Loss Components . . . 10
5.3 Computation . . . 10
5.4 Comparison with Traditional VaR . . . 11

6 Sobol Sensitivity Analysis **12**
6.1 Variance Decomposition . . . 12
6.2 First-Order and Total-Order Indices . . . 12
6.3 Saltelli Sampling Design . . . 13
6.4 Worked Example: Climate VaR Sensitivity . . . 13

7 TCFD Reporting Metrics **14**
7.1 Weighted Average Carbon Intensity (WACI) . . . 14
7.2 Financed Emissions . . . 14
7.3 Emission Scopes . . . 15
7.4 Portfolio Alignment: Implied Temperature Rise . . . 15
7.5 TCFD Reporting Template . . . 16

8 References **16**

1 Why Climate Risk Matters for Finance

Climate risk is no longer a reputational or ethical concern relegated to ESG departments. It is a *financial* risk — one that regulators, central banks, and standard-setters now treat as material to portfolio valuation, capital adequacy, and systemic stability.

1.1 The Regulatory Shift: From Voluntary to Mandatory

The Task Force on Climate-related Financial Disclosures (TCFD), established by the Financial Stability Board in 2017, laid the conceptual groundwork: firms should disclose how climate change affects their strategy, governance, risk management, and financial metrics. The TCFD framework was initially voluntary. That era is over.

- **ISSB (2023).** The International Sustainability Standards Board published IFRS S2 (*Climate-related Disclosures*), making TCFD-aligned reporting a baseline for capital markets globally. Jurisdictions including the UK, EU, Japan, and Australia have moved to adopt or mandate ISSB standards.
- **ECB (2025).** The European Central Bank's 2025 macroprudential stress test required Eurozone banks to model portfolio losses under NGFS climate scenarios, integrating transition and physical risk channels into supervisory capital assessments for the first time at scale.
- **EU CSRD (2024).** The Corporate Sustainability Reporting Directive requires approximately 50,000 companies to report under the European Sustainability Reporting Standards, which embed double materiality — both how climate affects the company and how the company affects climate.

Warning

The transition from voluntary to mandatory disclosure means that climate risk modeling is no longer optional for financial institutions. A bank that cannot quantify its portfolio exposure to a \$200/tCO_2 carbon price or a 3°C warming scenario faces both regulatory penalties and market repricing once peers begin disclosing.

1.2 Double Materiality

The concept of **double materiality** is central to the European framework and increasingly influential elsewhere:

Definition 1.1 (Double materiality)**.** A risk is **doubly material** if it operates in both directions:

1. **Outside-in (financial materiality):** Climate change affects the value of financial assets — through carbon pricing, physical damages, stranded assets, or liability risks.
2. **Inside-out (impact materiality):** The portfolio's financed activities contribute to climate change — through greenhouse gas emissions, deforestation, or resource depletion — creating systemic risk that feeds back into financial outcomes.

A coal-heavy portfolio is exposed from the outside-in (carbon prices reduce earnings) *and* contributes from the inside-out (financed emissions accelerate warming, increasing physical risk for all market participants).

1.3 The Scale of Exposure

The financial stakes are large:

- **Stranded assets.** An estimated $2.5 trillion in fossil fuel reserves become economically unviable if carbon prices rise to levels consistent with the Paris Agreement. These reserves are currently on corporate balance sheets, underpinning equity valuations and debt covenants.
- **Physical damages.** Under a 3°C+ warming pathway, the Nordhaus damage function implies cumulative GDP losses of 4% or more — concentrated in agriculture, insurance, real estate, and infrastructure.
- **Transition costs.** An orderly transition to net zero by 2050 requires annual clean-energy investment of approximately $4 trillion (IEA, 2023), implying massive capital reallocation away from carbon-intensive sectors.

Core Idea

Climate risk is not a single number. It is a *scenario-dependent* distribution of losses that varies with the speed and ambition of the low-carbon transition. An orderly transition imposes moderate, predictable transition costs but limits physical damages. A delayed or absent transition avoids near-term transition costs but exposes portfolios to severe, non-linear physical damages. The challenge for risk managers is to evaluate portfolios under *all* plausible pathways.

2 The NGFS Scenario Framework

The Network for Greening the Financial System (NGFS) provides the reference scenario set used by most central banks and supervisory authorities. Phase V (2024) defines six pathways organized along two axes: the speed and coordination of transition policy (orderly versus disorderly) and the level of physical risk (determined by the resulting warming outcome).

2.1 The Six Pathways

Table 1: NGFS Phase V scenario matrix. Temperature outcomes refer to global mean surface temperature increase by 2100 relative to pre-industrial levels. Shadow carbon prices are illustrative 2035 values in USD/tCO_2.

Scenario	**Transition Risk**	**Warming (°C)**	**Carbon Price ($/t$CO_2$)**	**Character**
Net Zero 2050	Orderly	1.5	∼300	Immediate, coordinated
Below 2°C	Orderly	1.7	∼200	Gradual strengthening
Low Demand	Orderly	1.6	∼150	Demand-side shift
Delayed Transition	Disorderly	1.8	∼350*	Late, abrupt policy
NDCs	Insufficient	2.6	∼50	Current pledges only
Current Policies	Minimal	3.0+	∼10	No new policy

*Delayed Transition carbon price is low until ∼2030, then spikes sharply — the disorderly character implies a sudden repricing event.

Key Insight

The six scenarios span a fundamental trade-off: **transition risk and physical risk are inversely correlated across pathways**. Orderly scenarios have high carbon prices (transition risk) but limit warming to 1.5–1.7°C (low physical risk). Current Policies has negligible transition risk but exposes portfolios to 3°C+ warming (high physical risk). The Delayed Transition scenario is particularly dangerous: it combines high transition risk (sudden policy action) with moderate physical risk (1.8°C warming already locked in).

2.2 Shadow Carbon Price Trajectories

The **shadow carbon price** is the implicit cost of carbon emissions required to achieve a given temperature target. Under the Net Zero 2050 pathway:

$$p_{\mathrm{CO}_2}(t) \approx \begin{cases} 50 + 50 \cdot (t - 2025) & \text{for } 2025 \leq t \leq 2030, \\ 300 + 20 \cdot (t - 2035) & \text{for } t > 2035, \end{cases} \tag{1}$$

reaching approximately \$300/tCO_2 by 2035 and continuing to rise thereafter. Under the Delayed Transition pathway, the carbon price remains below \$30/tCO_2 until approximately 2030, then rises sharply to \$350/tCO_2 by 2035 — a price shock that propagates through energy markets, credit spreads, and equity valuations simultaneously.

2.3 Short-Term Scenarios (May 2025)

NGFS introduced short-term scenarios with a 2025–2030 horizon, reflecting the recognition that long-run (2050–2100) pathways, while important for strategic planning, do not capture the near-term financial risks that drive supervisory stress testing and portfolio risk management. These short-term scenarios incorporate:

- Specific policy commitments (e.g., EU CBAM, US IRA provisions).
- Near-term commodity price dynamics and energy supply constraints.
- Technology readiness levels for key abatement options.

Go Deeper

The NGFS scenario portal (`https://www.ngfs.net/ngfs-scenarios-portal`) provides downloadable variable paths for GDP, carbon prices, energy mix, temperature, and sectoral output for all six pathways. These are the direct inputs to the Climate VaR computation in Section 5.

3 Transition Risk Modeling

Transition risk arises when the shift to a low-carbon economy changes the relative costs, revenues, and asset values of firms and sectors. The primary channel is carbon pricing — either explicit (carbon taxes, emissions trading systems) or implicit (regulation, shifts in demand).

3.1 Carbon Cost Propagation

The annualized carbon cost for a firm is:

$$C_{\text{carbon}} = \underbrace{e_i}_{\text{emission intensity}} \times \underbrace{p_{\text{CO}_2}}_{\text{carbon price}} \times \underbrace{R_i}_{\text{revenue}}, \tag{2}$$

where e_i is the firm's emission intensity measured in tonnes of CO_2 per million dollars of revenue (tCO_2/\$M), p_{CO_2} is the carbon price in dollars per tonne, and R_i is the firm's annual revenue.

Core Idea

Carbon cost scales *multiplicatively* in two policy-dependent variables: emission intensity (which firms can reduce through abatement) and carbon price (which is set by regulation or market forces). A firm with high emission intensity but low revenue may face less absolute carbon cost than a moderate-intensity firm with large revenue. Sector-level analysis must account for both.

3.2 Sector Emission Intensities

Emission intensities vary by orders of magnitude across sectors:

Table 2: Approximate Scope 1+2 emission intensities by sector. Values are representative and vary by geography and sub-industry.

Sector	**Intensity (tCO_2/\$M revenue)**	**Primary Source**
Energy (Oil & Gas)	∼500	Extraction, refining, combustion
Utilities	∼300	Power generation (coal, gas)
Materials (Steel, Cement)	∼200	Industrial processes
Transportation	∼150	Fuel combustion
Agriculture	∼120	Livestock, fertilizer, land use
Financials	∼50	Financed emissions (Scope 3)
Technology	∼10	Data centers, supply chain

3.3 Equity Repricing Under Carbon Costs

When a carbon cost is imposed, the firm's earnings decline. The first-order equity repricing is:

$$\Delta V_i = -\frac{C_{\text{carbon},i}}{\text{EBITDA}_i} \times V_i, \tag{3}$$

where V_i is the current equity value and EBITDA_i is earnings before interest, taxes, depreciation, and amortization. The ratio $C_{\text{carbon}}/\text{EBITDA}$ measures how much of the firm's earnings are consumed by carbon costs — a direct analogue of the interest coverage ratio for debt.

Definition 3.1 (Carbon earnings-at-risk)**.** The **carbon earnings-at-risk** for firm i under scenario s is:

$$\text{CarbonEaR}_{i,s} = \frac{e_i \cdot p_{\text{CO}_2}^{(s)} \cdot R_i}{\text{EBITDA}_i}. \tag{4}$$

A CarbonEaR above 1.0 implies that carbon costs exceed the firm's entire operating earnings — the firm becomes unprofitable under that carbon price.

3.4 Stranded Assets

Fossil fuel reserves are classified as **stranded** when the cost of extraction plus the carbon cost exceeds the market price of the commodity:

$$\text{Reserve is stranded} \iff c_{\text{extract}} + p_{\text{CO}_2} \cdot f_{\text{emission}} > p_{\text{market}}, \tag{5}$$

where c_{extract} is the per-barrel (or per-tonne) extraction cost, f_{emission} is the emission factor (tCO_2 per barrel), and p_{market} is the commodity price.

Example 3.1 (Stranded reserves under \$200/tCO_2)**.** Consider an oil producer with the following profile:

- Proven reserves: 2 billion barrels of oil equivalent (boe).
- Extraction cost distribution: 40% of reserves at \$30/boe, 35% at \$50/boe, 25% at \$70/boe.
- Emission factor: 0.43 tCO_2/boe (industry average).
- Market oil price: \$75/boe.

Under a carbon price of \$200/tCO_2, the effective carbon cost per barrel is $200 \times 0.43 =$ \$86/boe. The total cost (extraction + carbon) for each reserve tranche:

Tranche	Extraction	Total Cost	Status
40% at \$30/boe	\$30	\$30 + \$86 = \$116	Stranded (> \$75)
35% at \$50/boe	\$50	\$50 + \$86 = \$136	Stranded (> \$75)
25% at \$70/boe	\$70	\$70 + \$86 = \$156	Stranded (> \$75)

At \$200/tCO_2, **100%** of this producer's reserves are stranded — none can be profitably extracted. Even at a more moderate \$100/tCO_2 (carbon cost = \$43/boe), the highest-cost tranche (\$70 + \$43 = \$113) and mid-cost tranche (\$50 + \$43 = \$93) are stranded, leaving only 40% of reserves viable. The equity write-down from stranding 60% of reserves at \$100/tCO_2 is:

$$\Delta V = -0.60 \times \text{NPV(reserves)} \times \frac{V_{\text{equity}}}{\text{EV}}, \tag{6}$$

where EV is enterprise value. For a firm where reserves represent 70% of enterprise value, this implies a ∼42% equity loss.

Warning

Stranded asset risk is highly non-linear. At low carbon prices, most reserves remain viable. Above a critical threshold — which varies by extraction cost profile — the fraction of stranded reserves jumps sharply. This creates *cliff risk*: a small increase in carbon price near the threshold can trigger a large write-down.

4 Physical Risk Modeling

Physical risk is the direct economic damage from climate change. It operates through two channels: **acute** risks (discrete extreme events whose frequency and severity increase with warming) and **chronic** risks (gradual shifts that erode productivity and asset values over decades).

4.1 The Nordhaus Damage Function

The workhorse model for aggregate physical damages is the quadratic damage function introduced by Nordhaus (2017) in the DICE model:

$$D(T) = \alpha \cdot T^2, \tag{7}$$

where $D(T)$ is the fraction of GDP lost at a global mean temperature increase of T degrees Celsius above pre-industrial levels, and α is a calibration parameter. Nordhaus calibrates $\alpha \approx 0.00267$, implying:

Table 3: GDP loss under the Nordhaus quadratic damage function ($\alpha = 0.00267$).

Warming (°C)	**GDP Loss (%)**	**NGFS Scenario**
1.5	0.6%	Net Zero 2050
2.0	1.1%	Below 2°C
2.5	1.7%	—
3.0	2.4%	Current Policies
4.0	4.3%	Current Policies (high sensitivity)

Core Idea

The quadratic functional form is crucial: damages are **convex** in temperature. The marginal damage of each additional degree of warming is *increasing*. Moving from 2°C to 3°C is more than twice as costly as moving from 1°C to 2°C. This convexity is the fundamental reason why aggressive mitigation (avoiding high warming) is economically rational even when discounted at market rates.

Remark 4.1. The Nordhaus damage function has been criticized as too conservative. Howard and Sterner (2017) survey the empirical literature and find that including non-market damages (biodiversity loss, health impacts, conflict) and accounting for persistent growth effects roughly doubles the damage coefficient to $\alpha \approx 0.005$, implying ∼8% GDP loss at 4°C. Our framework treats α as a parameter to be stress-tested rather than a fixed constant.

4.2 Acute Physical Risk: Extreme Weather Events

Acute physical risk manifests as discrete, high-severity events: floods, hurricanes, wildfires, and heat waves. Climate change increases both the frequency and the severity of these events, shifting the tail of the loss distribution.

Definition 4.1 (Climate-adjusted extreme value distribution)**.** Let L denote the annual maximum loss from a climate-sensitive peril (e.g., flood damage). Under the Generalized Pareto Distribution (GPD) framework from extreme value theory (see Project 05), exceedances above a threshold u follow:

$$\mathbb{P}(L - u > y \mid L > u) = \left(1 + \xi \frac{y}{\sigma}\right)^{-1/\xi}, \quad y > 0, \tag{8}$$

where $\xi > 0$ is the shape parameter (controlling tail heaviness) and $\sigma > 0$ is the scale parameter. Under climate change, both parameters are functions of temperature:

$$\sigma(T) = \sigma_0 \cdot \exp(\beta_\sigma \cdot T), \qquad \xi(T) = \xi_0 + \beta_\xi \cdot T, \tag{9}$$

reflecting that warming increases both the scale (larger typical extreme losses) and the shape (heavier tails — more "black swan" events).

Example 4.1 (100-year flood loss under warming)**.** Consider flood losses for a coastal property portfolio. Under current climate ($T = 1.1$°C above pre-industrial):

- Threshold: $u = \$50$M (losses exceeding this are "extreme").
- GPD parameters: $\sigma_0 = 20$, $\xi_0 = 0.15$.

- 100-year return level (0.01 annual exceedance probability):

$$L_{100} = u + \frac{\sigma_0}{\xi_0}\left[\left(\frac{1}{0.01}\right)^{\xi_0} - 1\right] = 50 + \frac{20}{0.15}\left[100^{0.15} - 1\right] = 50 + 133.3 \times 0.995 = \$183\text{M}. \quad (10)$$

Under 3°C warming, with $\beta_\sigma = 0.08$ and $\beta_\xi = 0.01$ per degree:

- $\sigma(3.0) = 20 \cdot \exp(0.08 \times 3.0) = 20 \times 1.271 = 25.4$.
- $\xi(3.0) = 0.15 + 0.01 \times 3.0 = 0.18$.

$$L_{100}^{3°\text{C}} = 50 + \frac{25.4}{0.18}\left[100^{0.18} - 1\right] = 50 + 141.1 \times 1.318 = \$236\text{M}. \quad (11)$$

The 100-year flood loss increases from \$183M to \$236M — a **29% increase** — under 3°C warming. More critically, what was a 100-year event under current climate becomes approximately a 50-year event, doubling the expected annual frequency.

4.3 Chronic Physical Risk

Chronic risks accumulate gradually and affect asset productivity and valuations over decades:

- **Sea-level rise.** Coastal real estate and infrastructure face permanent inundation or increased flood insurance costs. Under 3°C, global mean sea level rises approximately 0.5–1.0m by 2100.
- **Heat stress.** Labor productivity declines as the number of days exceeding safe working temperatures increases. Outdoor sectors (agriculture, construction, mining) are most exposed.
- **Agricultural yield decline.** Crop yields decline non-linearly with temperature beyond optimal growing ranges. Each degree above 2°C reduces global yields by approximately 5–10% for staple crops (IPCC AR6).
- **Water stress.** Changing precipitation patterns affect hydropower, agriculture, and industrial water supply.

4.4 Sector Exposure to Physical Risk

Table 4: Sector exposure to physical risk channels. H = High, M = Medium, L = Low.

Sector	Acute (Floods/Storms)	Sea-Level Rise	Heat Stress	Yield Decline
Agriculture	H	M	H	H
Insurance	H	H	M	L
Real Estate	H	H	M	L
Infrastructure	H	H	M	L
Utilities	M	M	M	L
Transportation	M	M	L	L
Technology	L	L	L	L

Key Insight

Transition risk and physical risk affect *different* sectors. Energy, utilities, and materials face the highest transition risk (carbon intensity). Agriculture, insurance, and real estate face the highest physical risk (exposure to weather and temperature). A well-diversified portfolio is exposed to *both* channels simultaneously, and the correlation between them depends on the scenario pathway.

5 Climate Value-at-Risk

Climate VaR aggregates transition and physical losses into a single portfolio-level risk metric, evaluated under each NGFS scenario.

5.1 Definition

Definition 5.1 (Climate Value-at-Risk)**.** For a portfolio with N holdings, under NGFS scenario s with time horizon H (typically 10–30 years), the **Climate VaR** at confidence level $1-\alpha$ is:

$$\text{ClimateVaR}_\alpha^{(s)} = \inf\left\{ l \geq 0 : \mathbb{P}\left(\sum_{i=1}^{N} w_i \cdot \left(L_i^{\text{trans}}(s) + L_i^{\text{phys}}(s) \right) \leq l \right) \geq 1-\alpha \right\}, \tag{12}$$

where w_i is the portfolio weight, $L_i^{\text{trans}}(s)$ is the transition loss for holding i under scenario s, and $L_i^{\text{phys}}(s)$ is the physical loss.

5.2 Loss Components

The total loss for holding i under scenario s is:

$$L_i^{(s)} = L_i^{\text{trans}}(s) + L_i^{\text{phys}}(s), \tag{13}$$

where:

Transition loss.

$$L_i^{\text{trans}}(s) = \frac{e_i \cdot p_{\text{CO}_2}^{(s)}(H) \cdot R_i}{\text{EBITDA}_i}, \tag{14}$$

expressed as a fraction of current equity value (i.e., the carbon earnings-at-risk from Definition 3.1).

Physical loss.

$$L_i^{\text{phys}}(s) = \phi_i \cdot D\Big(T^{(s)}(H)\Big) = \phi_i \cdot \alpha \cdot \big(T^{(s)}(H)\big)^2, \tag{15}$$

where $\phi_i \in [0,1]$ is the physical risk exposure factor for holding i (from Table 4), $T^{(s)}(H)$ is the cumulative warming by horizon H under scenario s, and $D(\cdot)$ is the Nordhaus damage function.

5.3 Computation

For deterministic scenarios (a single pathway per NGFS scenario), the Climate VaR reduces to a point estimate:

$$\text{ClimateVaR}^{(s)} = \sum_{i=1}^{N} w_i \cdot L_i^{(s)}. \tag{16}$$

For stochastic scenarios (where carbon prices, temperatures, and damages are random variables within each pathway), we simulate:

Algorithm 1: Stochastic Climate VaR

Input: Portfolio weights $\{w_i\}$, firm data $\{e_i, R_i, \text{EBITDA}_i, \phi_i\}$, NGFS scenario s, number of simulations M, confidence level $1-\alpha$.

1. For $m = 1, \ldots, M$:
   (a) Draw carbon price: $p_{\text{CO}_2}^{(m)} \sim F_{\text{CO}_2}^{(s)}$.
   (b) Draw temperature: $T^{(m)} \sim F_T^{(s)}$.
   (c) Draw GDP growth path: $g^{(m)} \sim F_g^{(s)}$.
   (d) Compute transition loss for each holding: $L_i^{\text{trans},(m)} = e_i \cdot p_{\text{CO}_2}^{(m)} \cdot R_i/\text{EBITDA}_i$.
   (e) Compute physical loss for each holding: $L_i^{\text{phys},(m)} = \phi_i \cdot \alpha \cdot (T^{(m)})^2$.
   (f) Compute portfolio loss: $L^{(m)} = \sum_{i=1}^{N} w_i \cdot (L_i^{\text{trans},(m)} + L_i^{\text{phys},(m)})$.
2. Sort $\{L^{(1)}, \ldots, L^{(M)}\}$.
3. Return $\text{ClimateVaR}_\alpha^{(s)} = L^{(\lceil M(1-\alpha) \rceil)}$.

5.4 Comparison with Traditional VaR

Table 5: Traditional VaR vs. Climate VaR.

Dimension	**Traditional VaR**	**Climate VaR**
Time horizon	1 day – 10 days	5 – 30 years
Risk drivers	Market prices, volatility	Carbon price, temperature, policy
Data source	Historical returns	NGFS scenario projections
Distribution	Estimated from data	Scenario-conditioned simulation
Stationarity	Assumed (or windowed)	Explicitly non-stationary
Interpretation	"Loss in normal conditions"	"Loss under pathway s"

Example 5.1 (Portfolio Climate VaR across pathways)**.** Consider a portfolio with three sector exposures:

- 30% Energy ($e = 500$, EBITDA margin = 15%, $\phi = 0.3$).
- 40% Financials ($e = 50$, EBITDA margin = 35%, $\phi = 0.4$).
- 30% Technology ($e = 10$, EBITDA margin = 25%, $\phi = 0.1$).

Under the Delayed Transition pathway ($p_{\text{CO}_2} = \$350$, $T = 1.8°\text{C}$ by 2050):

$$L_{\text{Energy}}^{\text{trans}} = \frac{500 \times 350 \times 10^{-6}}{0.15} = 1.167 \quad \text{(i.e., 116.7\% of equity)}, \tag{17}$$

$$L_{\text{Fin}}^{\text{trans}} = \frac{50 \times 350 \times 10^{-6}}{0.35} = 0.050 \quad (5.0\%), \tag{18}$$

$$L_{\text{Tech}}^{\text{trans}} = \frac{10 \times 350 \times 10^{-6}}{0.25} = 0.014 \quad (1.4\%). \tag{19}$$

Physical losses (Nordhaus, $\alpha = 0.00267$, $T = 1.8$): $D(1.8) = 0.00267 \times 1.8^2 = 0.87\%$.

$$L^{\text{phys}}_{\text{Energy}} = 0.3 \times 0.0087 = 0.26\%, \tag{20}$$

$$L^{\text{phys}}_{\text{Fin}} = 0.4 \times 0.0087 = 0.35\%, \tag{21}$$

$$L^{\text{phys}}_{\text{Tech}} = 0.1 \times 0.0087 = 0.09\%. \tag{22}$$

Portfolio Climate VaR:

$$\text{ClimateVaR} = 0.30 \times 1.170 + 0.40 \times 0.054 + 0.30 \times 0.015 = 37.7\%. \tag{23}$$

Under the delayed transition pathway, this portfolio could lose approximately 38% of its value over the projection horizon — driven almost entirely by the energy sector's transition risk.

6 Sobol Sensitivity Analysis

Climate VaR depends on multiple uncertain inputs: carbon price, temperature, GDP growth, discount rates, emission intensities, and physical exposure factors. **Sobol sensitivity analysis** provides a rigorous, variance-based decomposition of which inputs drive the most uncertainty in the output.

6.1 Variance Decomposition

Let $Y = f(X_1, X_2, \ldots, X_k)$ be the Climate VaR as a function of k uncertain inputs. The total variance of Y can be decomposed (under independence of inputs) as:

$$\text{Var}(Y) = \sum_{i=1}^{k} V_i + \sum_{i<j} V_{ij} + \cdots + V_{1,2,\ldots,k}, \tag{24}$$

where $V_i = \text{Var}\big(\mathbb{E}[Y \mid X_i]\big)$ is the variance attributable to X_i alone, V_{ij} captures the interaction between X_i and X_j, and so on.

6.2 First-Order and Total-Order Indices

Definition 6.1 (Sobol indices)**.** The **first-order Sobol index** for input X_i is:

$$S_i = \frac{V_i}{\text{Var}(Y)} = \frac{\text{Var}\big(\mathbb{E}[Y \mid X_i]\big)}{\text{Var}(Y)}. \tag{25}$$

It measures the fraction of output variance attributable to X_i *alone*, averaging over all other inputs.

The **total-order Sobol index** for input X_i is:

$$S_{T_i} = 1 - \frac{\text{Var}\big(\mathbb{E}[Y \mid X_{\sim i}]\big)}{\text{Var}(Y)} = \frac{\mathbb{E}\big[\text{Var}(Y \mid X_{\sim i})\big]}{\text{Var}(Y)}, \tag{26}$$

where $X_{\sim i}$ denotes all inputs *except* X_i. The total-order index captures the variance attributable to X_i including all interactions with other inputs.

Core Idea

S_i measures the *main effect* of input i: how much output variance would be eliminated if X_i were fixed at its true value. S_{T_i} measures the main effect *plus all interactions*: fixing X_i eliminates this fraction of output variance. The difference $S_{T_i} - S_i$ quantifies the importance of interactions involving X_i. If all indices are close to their first-order values ($S_{T_i} \approx S_i$ for all i), the model is approximately additive and interactions are negligible.

6.3 Saltelli Sampling Design

Efficient estimation of Sobol indices uses the Saltelli (2002) sampling design. For k inputs and N base samples, this requires $N(k+2)$ model evaluations:

Algorithm 2: Saltelli Sampling for Sobol Indices

Input: Number of base samples N, number of inputs k, input distributions $\{F_1, \ldots, F_k\}$.

1. Generate two independent $N \times k$ sample matrices $\boldsymbol{A}$ and $\boldsymbol{B}$, where each column is drawn from the corresponding input distribution.
2. For each input $i = 1, \ldots, k$, construct matrix $\boldsymbol{A}_B^{(i)}$: take $\boldsymbol{A}$ and replace column i with column i from $\boldsymbol{B}$.
3. Evaluate the model at all $N(k+2)$ points: $f(\boldsymbol{A})$, $f(\boldsymbol{B})$, and $f(\boldsymbol{A}_B^{(i)})$ for each i.
4. Estimate first-order indices:
$$\hat{S}_i = \frac{\frac{1}{N}\sum_{j=1}^{N} f(\boldsymbol{B})_j \big(f(\boldsymbol{A}_B^{(i)})_j - f(\boldsymbol{A})_j\big)}{\mathrm{Var}(Y)}. \tag{27}$$
5. Estimate total-order indices:
$$\hat{S}_{T_i} = \frac{\frac{1}{2N}\sum_{j=1}^{N} \big(f(\boldsymbol{A})_j - f(\boldsymbol{A}_B^{(i)})_j\big)^2}{\mathrm{Var}(Y)}. \tag{28}$$

6.4 Worked Example: Climate VaR Sensitivity

Example 6.1 (Sobol decomposition of Climate VaR)**.** We apply Sobol analysis to the stochastic Climate VaR model (Algorithm 5.3) with five uncertain inputs:

Input	**Distribution**	S_i	S_{T_i}	$S_{T_i} - S_i$
Carbon price (p_{CO_2})	LogNormal(μ, σ)	0.45	0.52	0.07
Temperature (T)	Normal(μ_T, σ_T)	0.25	0.30	0.05
GDP growth (g)	Normal(μ_g, σ_g)	0.15	0.18	0.03
Emission intensity (e)	Uniform($0.8e_0$, $1.2e_0$)	0.08	0.10	0.02
Discount rate (r)	Uniform(2%, 5%)	0.04	0.05	0.01
Sum		0.97	1.15	

Interpretation:

- **Carbon price** is the dominant driver, accounting for 45% of output variance on its own and 52% including interactions. Policy uncertainty about the trajectory and level of carbon pricing dominates the Climate VaR uncertainty.
- **Temperature** is the second-largest driver (25%), primarily affecting the physical risk channel through the quadratic damage function.
- **GDP growth** contributes 15%, affecting both revenue projections and the denominator of the damage function.
- The sum of first-order indices ($\sum S_i = 0.97$) is close to 1, indicating that the model is approximately *additive* — interactions between inputs are modest.

- The sum of total-order indices exceeds 1 ($\sum S_{T_i} = 1.15$) because total-order indices double-count shared interaction variance.

Key Insight

Sobol analysis reveals that **carbon pricing policy** is the single most important driver of climate portfolio risk — more important than the physical temperature outcome. This finding has a direct implication: risk managers should focus scenario analysis on the range of plausible carbon price trajectories and treat physical risk as a secondary (but still material) driver. For portfolios with high physical exposure (real estate, agriculture), the ranking may reverse.

7 TCFD Reporting Metrics

The TCFD framework (and its successor, ISSB IFRS S2) defines specific metrics for disclosing portfolio-level climate exposure. These metrics complement the scenario analysis in Sections 3–5 with standardized, comparable measures.

7.1 Weighted Average Carbon Intensity (WACI)

Definition 7.1 (WACI)**.** The **Weighted Average Carbon Intensity** of a portfolio is:

$$\text{WACI} = \sum_{i=1}^{N} w_i \cdot \text{CI}_i, \tag{29}$$

where w_i is the portfolio weight of holding i and CI_i is the carbon intensity of the issuer (tCO_2/\$M revenue, Scope 1+2).

WACI is the TCFD's recommended cross-sector metric because it:

- Normalizes by revenue, making firms of different sizes comparable.
- Uses portfolio weights, reflecting actual exposure.
- Is readily computable from publicly available emissions data.

Example 7.1 (Portfolio WACI computation)**.**

Holding	**Weight** (w_i)	**CI**$_i$ **(tCO**$_2$**/\$M)**	$w_i\times$ **CI**$_i$
Oil Major A	15%	480	72.0
Utility B	10%	310	31.0
Steel Co C	10%	220	22.0
Bank D	25%	45	11.3
Tech E	20%	8	1.6
Healthcare F	20%	15	3.0
Portfolio WACI			**140.9**

The portfolio's WACI of 140.9 tCO_2/\$M revenue can be benchmarked against the MSCI ACWI index (∼130 tCO_2/\$M) or a Paris-aligned benchmark (∼70 tCO_2/\$M).

7.2 Financed Emissions

While WACI measures *intensity*, financed emissions measure *absolute* contribution to global emissions:

Definition 7.2 (Financed emissions)**.** The **financed emissions** attributable to a portfolio are:

$$E_{\text{financed}} = \sum_{i=1}^{N} \underbrace{\frac{V_i}{\text{EVIC}_i}}_{\text{ownership share}} \times \underbrace{E_i}_{\text{company emissions}}, \tag{30}$$

where V_i is the portfolio's investment in holding i, EVIC_i is the enterprise value including cash of the issuer, and E_i is the issuer's total Scope 1+2 emissions (tCO_2).

7.3 Emission Scopes

Corporate greenhouse gas emissions are classified into three scopes under the GHG Protocol:

Table 6: GHG Protocol emission scopes.

Scope	**Description**	**Data Quality**
Scope 1	Direct emissions from owned or controlled sources (e.g., combustion in facilities, company vehicles).	High
Scope 2	Indirect emissions from purchased electricity, steam, heating, and cooling.	High
Scope 3	All other indirect emissions in the value chain — upstream (purchased goods, business travel) and downstream (use of sold products, end-of-life treatment).	Low (estimated)

Warning

For many sectors, Scope 3 emissions dwarf Scope 1+2. An oil company's Scope 1+2 covers extraction and refining; its Scope 3 includes the combustion of all sold fuel — typically 5–10× larger. A bank's Scope 1+2 is negligible (office buildings), but its Scope 3 (financed emissions from its loan book) is enormous. Reporting only Scope 1+2 can drastically understate a portfolio's true climate footprint. ISSB IFRS S2 requires Scope 3 disclosure "to the extent that it is material" — in practice, this means most large firms must report it.

7.4 Portfolio Alignment: Implied Temperature Rise

The most forward-looking TCFD metric is the portfolio's **implied temperature rise** (ITR):

Definition 7.3 (Implied Temperature Rise)**.** The ITR of a portfolio estimates the global warming that would result if the entire economy had the same emission trajectory as the portfolio's holdings. It is computed by:

1. Projecting each holding's emissions through 2050 based on its disclosed targets and historical trajectory.
2. Aggregating to a portfolio-level emission pathway.
3. Mapping the pathway to a temperature outcome using a carbon budget model.

A portfolio aligned with the Paris Agreement should have an ITR $\leq 1.5°\text{C}$. Portfolios with high fossil fuel exposure typically have ITR values of 2.5–3.5°C.

7.5 TCFD Reporting Template

A complete TCFD-aligned climate risk report includes:

TCFD Disclosure Template

1. **Governance:** Board oversight of climate risk; management's role in assessing and managing climate-related risks.
2. **Strategy:** Climate-related risks and opportunities over short, medium, and long term; impact on business, strategy, and financial planning; resilience under different climate scenarios.
3. **Risk Management:** Processes for identifying, assessing, and managing climate-related risks; integration into overall risk management.
4. **Metrics and Targets:**
   - WACI (Eq. 29): _____ tCO_2/$M revenue.
   - Financed emissions (Eq. 30): _____ tCO_2.
   - Scope breakdown: Scope 1 _____, Scope 2 _____, Scope 3 _____.
   - Climate VaR by scenario (Eq. 12):
     - Net Zero 2050: _____%
     - Below 2°C: _____%
     - Delayed Transition: _____%
     - Current Policies: _____%
   - Implied Temperature Rise: _____°C.
   - Reduction targets: _____.

Read Before Coding

This section connects directly to the code implementation. The pipeline must: (1) ingest firm-level emission data and carbon price paths from NGFS; (2) compute WACI and financed emissions at the portfolio level; (3) run the Climate VaR simulation (Algorithm 5.3) for each scenario; (4) perform Sobol sensitivity analysis (Algorithm 6.3); (5) produce a TCFD-formatted output with all metrics populated.

8 References

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